

SQL Project on online book store



Hi, I'm Amaan Ali, a BSc final year student and aspiring Data Analyst passionate about data analytics and technology. I am currently learning SQL, Excel, Power BI, and data visualization to build strong analytical skills and solve real-world business problems.





Introduction

This project is based on an Online Book Store database management system using SQL. The project includes three connected datasets: Books, Customers, and Orders, which were used to perform data analysis and solve real-world business queries.

The main objective of this project was to analyze book sales, customer behavior, stock management, and revenue generation using SQL queries. The project covers both basic and advanced SQL operations such as JOIN, GROUP BY, aggregate functions, filtering, sorting, and subqueries.



**1) Retrieve all books in the
"Fiction" genre**

```
select * from books  
where genre='fiction';
```

**2) Find books published
after the year 1950**

```
select * from books  
where Published_Year>1950;
```

**3) List all customers from
the Canada**

```
select * from customer  
where country = "canada";
```

**4) Show orders placed in
November 2023**

```
select * from orders  
where Order_Date BETWEEN  
"2023-11-1" AND "2023-11-30";
```

5) Retrieve the total stock of books available

```
SELECT sum(stock) as toal_stock  
from books;
```

6) Find the details of the most expensive book

```
SELECT * FROM books  
ORDER BY price DESC  
LIMIT 1;
```

7) Show all customers who ordered more than 1 quantity of a book

```
select * from orders  
where Quantity>1;
```

8) Retrieve all orders where the total amount exceeds \$20

```
select * from orders  
where Total_Amount>20;
```

9) List all genres available in the Books table

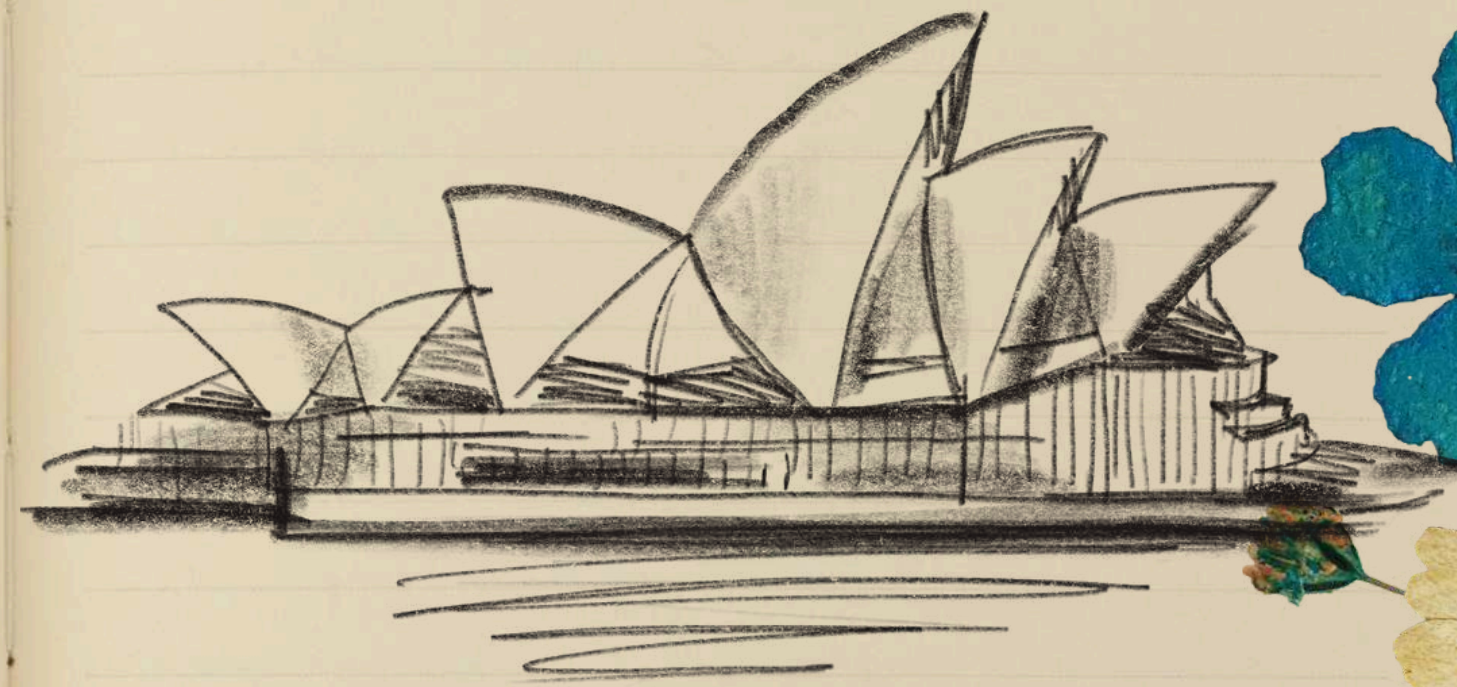
select distinct genre
from books;

10) Find the book with the lowest stock

select * from books
order by stock asc;

11) Calculate the total revenue generated from all orders

SELECT
SUM(total_amount) AS Revenue
FROM
orders;



Advance sql squery

1) Retrieve the total number of books sold for each genre

```
SELECT b.genre, SUM(o.Quantity)  
AS Total_Books_Sold  
FROM orders o  
JOIN books b ON  
o.book_id = b.book_id  
GROUP BY b.Genre;
```

2) Find the average price of books in the "Fantasy" genre

```
select avg(price) as average_price  
from books  
where genre ="fantasy";
```

3) List customers who have placed
at least 2 orders
`select customer_ID, count(order_ID)`
`AS Order_Count`
`from orders`
`group by customer_ID`
`having count(order_ID) >=2;`

4) Find the most frequently ordered
book
`select o.book_id, b.title, count(o.order_id)`
`as Order_Count`
`from orders o`
`join books b on o.book_id=b.book_id`
`group by o.book_id , b.title`
`order by order_count desc limit 1;`

5) Show the top 3 most expensive
books of 'Fantasy' Genre

`select * from books`
`where genre ="fantasy"`
`order by price desc limit 3;`

6) Retrieve the total quantity of
books sold by each author

`SELECT`
`b.author, SUM(o.quantity) AS Total_Books_Sold`
`FROM orders o`
`JOIN books b ON o.book_id = b.book_id`
`GROUP BY b.author;`

7) List the cities where customers who spent over \$30 are located

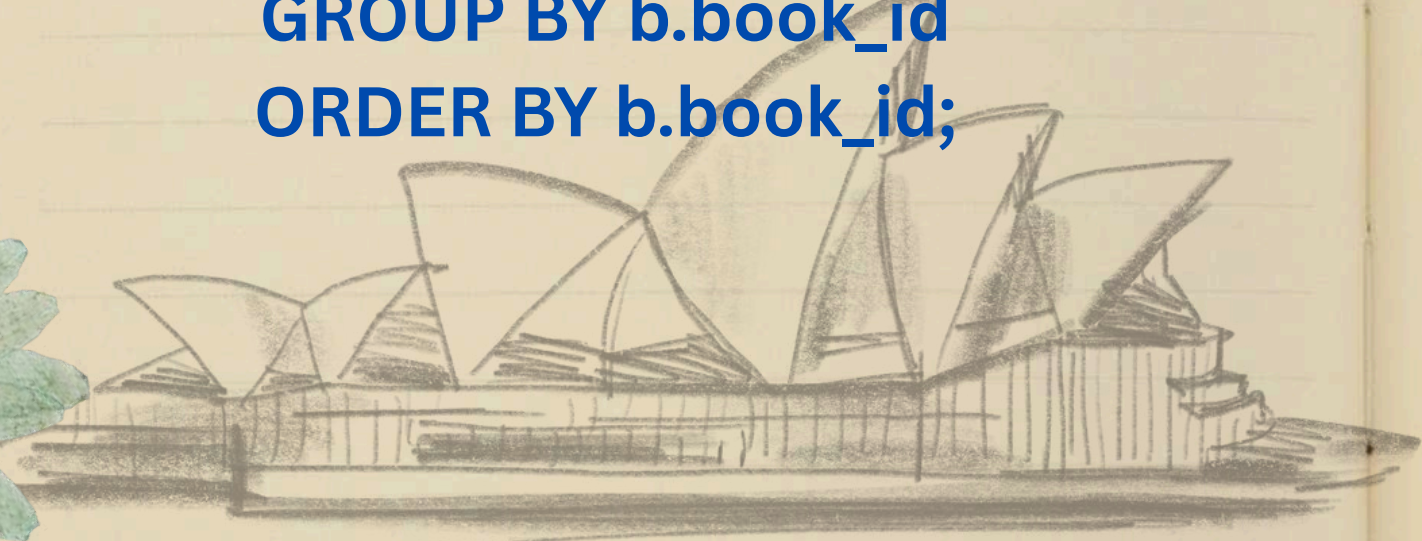
```
SELECT DISTINCT  
c.city, total_amount  
FROM orders o  
JOIN onlinebookstore.customer c  
ON o.customer_id = c.customer_id  
WHERE o.total_amount > 30;
```

8) Find the customer who spent the most on orders

```
SELECT  
c.customer_id, c.name, SUM(o.total_amount)  
AS Total_Spent  
FROM orders o  
JOIN onlinebookstore.customer c  
ON o.customer_id = c.customer_id  
GROUP BY c.customer_id , c.name  
ORDER BY Total_Spent DESC  
LIMIT 1;
```

9) Calculate the stock remaining after fulfilling all orders

```
SELECT
    b.book_id, b.title, b.stock,
    COALESCE(SUM(o.quantity), 0)
    AS order_quantity,
    b.stock - COALESCE(SUM(o.quantity), 0)
    AS remaining_quantity
FROM books b LEFT JOIN orders o
ON b.book_id = o.book_id
GROUP BY b.book_id
ORDER BY b.book_id;
```



Thank You

Through this project, I solved multiple business problems like finding the most frequently ordered books, calculating total revenue, identifying top customers, analyzing genre-wise sales, and checking remaining stock after orders.

Amaan Ali